RMIT Classification: Trusted

OENG1209: Engineering Science

Week 2:

Kinematics

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Today's Topics

- **Motion in One Dimension**
  - Velocity, acceleration
  - Motion with constant acceleration
  - Freely falling motion
- **Motion in Two Dimensions**
  - Projectile motion
  - Relative motion

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1. Motion in One Dimension

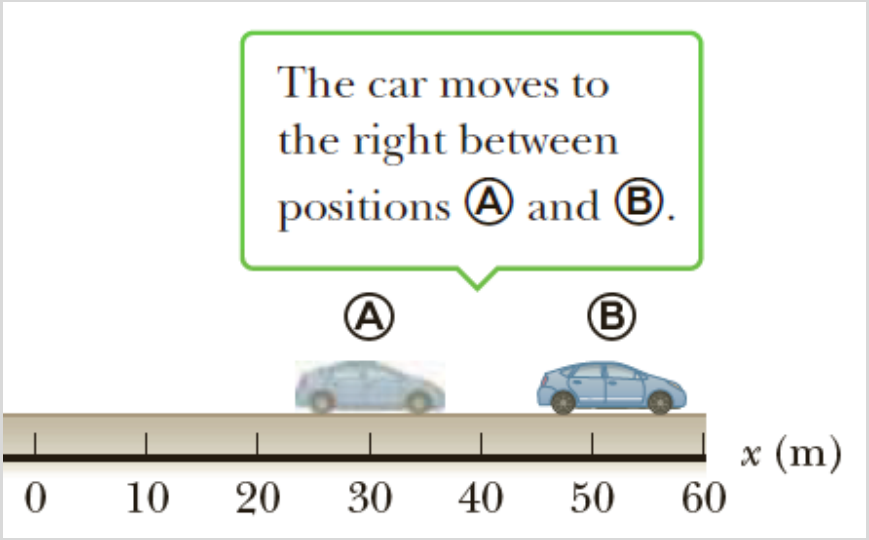
Motion in a straight-line

Motion in one dimension

Displacement

The displacement Δx of an object is defined as its change in position and is given by

$$\Delta x = x_f - x_i$$

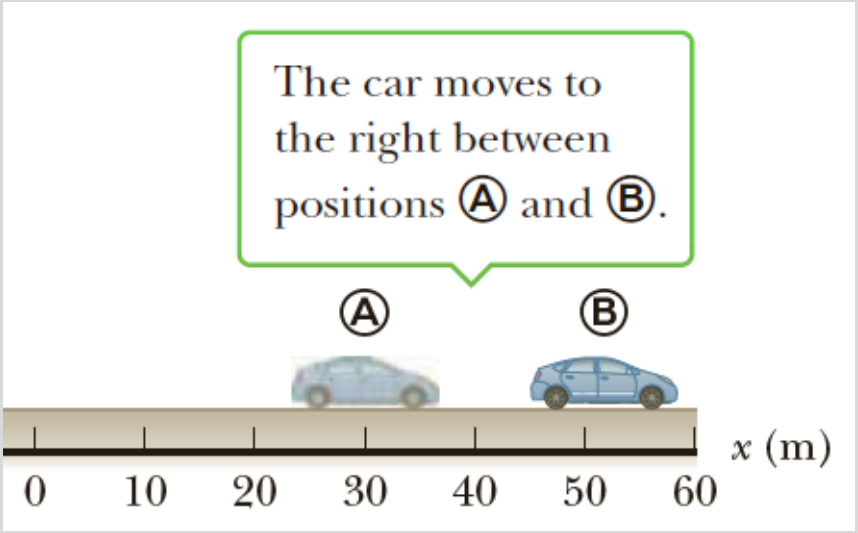


Motion in one dimension

Average velocity

The average velocity $\bar{v}$ during a time interval Δt is the displacement Δx divided by Δt

$$\bar{v} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$



Motion in one dimension

Average velocity – graphical interpretation

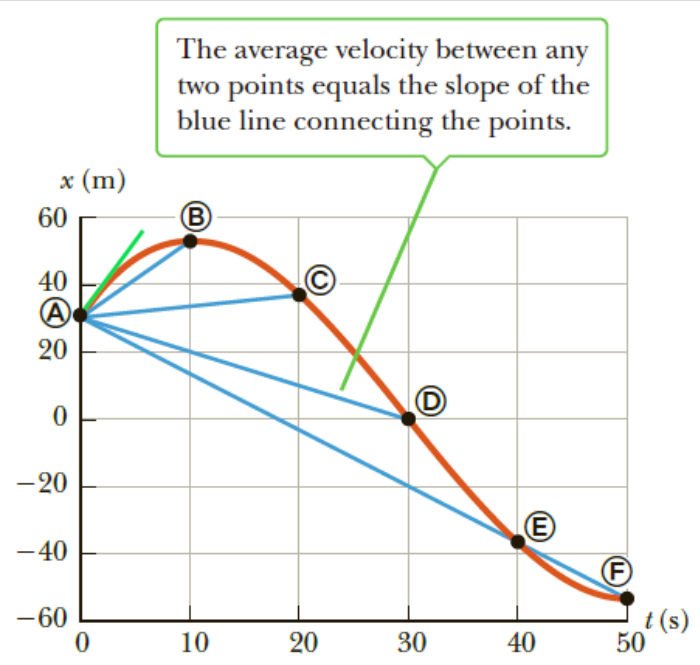
The slope is defined as

sequence line
connect 2 points

avg x

$$\text{slope} = \frac{\text{change in vertical axis}}{\text{change in horizontal axis}} \quad \Delta y \quad \Delta x \quad \text{or} \quad \frac{\Delta x}{\Delta t}$$

Position	t (s)	x (m)
Ⓐ	0	30
Ⓑ	10	52
Ⓒ	20	38
Ⓓ	30	0
Ⓔ	40	−37
Ⓕ	50	−53



The average velocity is the slope of the secant line.

Motion in one dimension
Instantaneous velocity

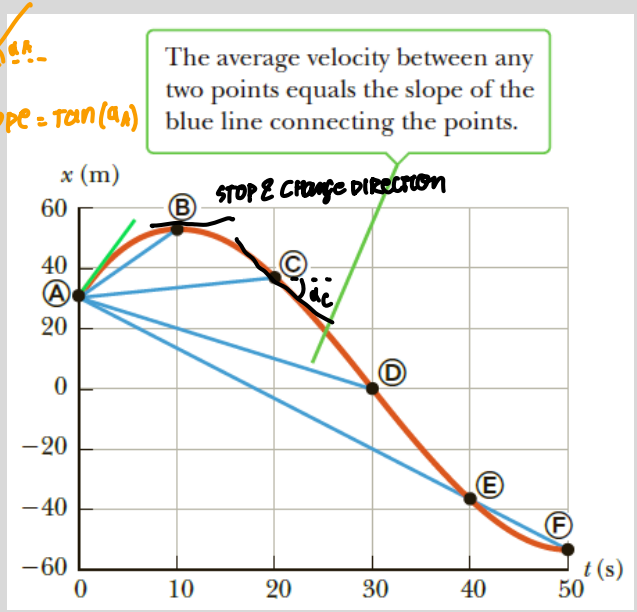
Average velocity doesn't take into account the details of what happens during an interval of time. The instantaneous velocity v is the limit of the average velocity as the time interval Δt becomes infinitesimally small

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

The instantaneous velocity is the slope of the tangent line.

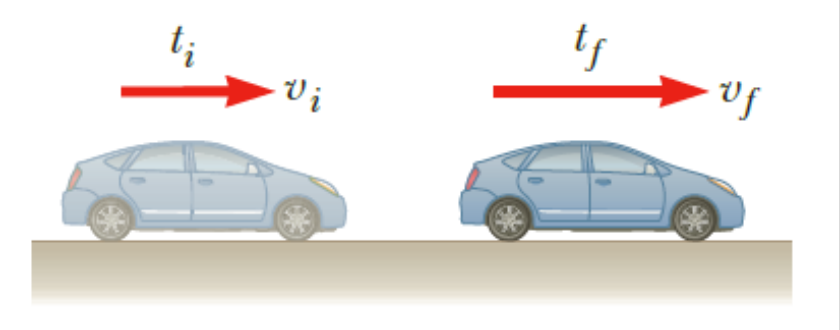
POINT B -> CLOSER TO A -> TANGENT LINE

SLOPE OF TANGENT LINE OF POINT ...



Motion in one dimension

Acceleration (velocity can change)



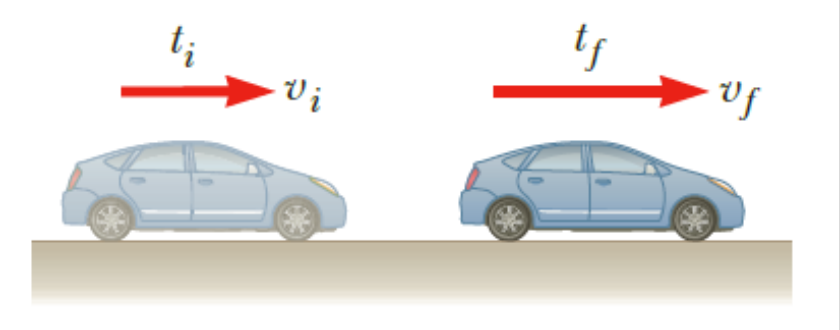
- Step harder on the gas pedal → increase the velocity
- Apply the brakes → decrease the velocity

Definition: The **average** acceleration change in velocity over change in time

$$\bar{a} \equiv \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$$

Motion in one dimension

Instantaneous acceleration



Definition: The **instantaneous** acceleration

ACCELERATION IS THE DERIVERTIVE OF ACCELERATION

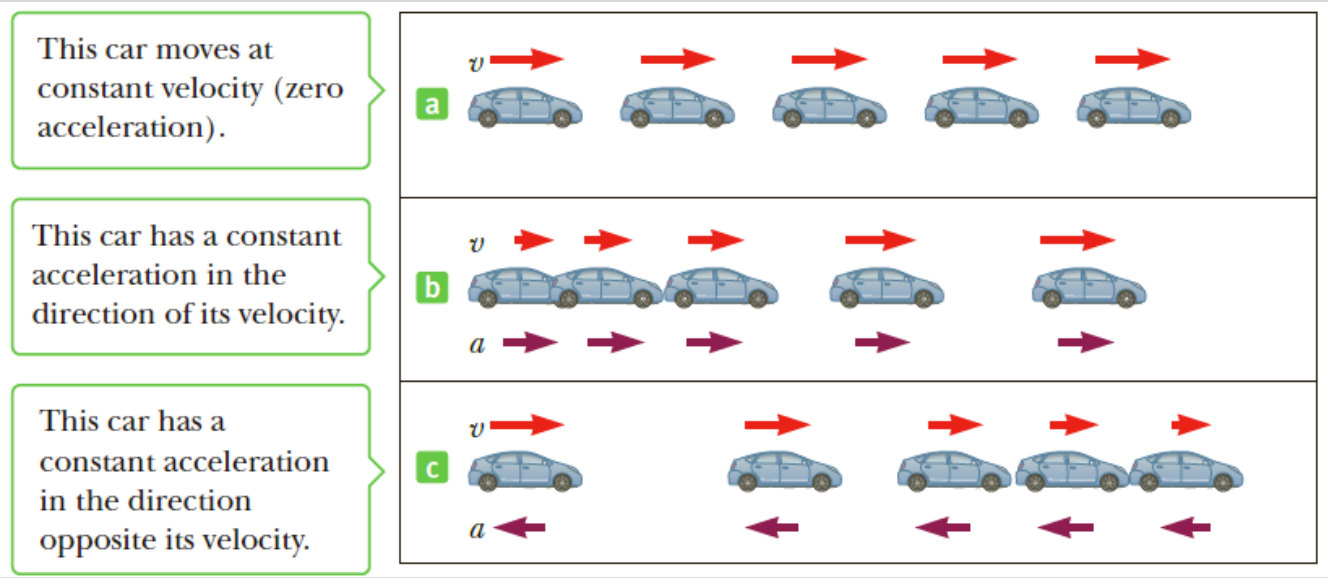
$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

Motion in one dimension

Acceleration illustration

The **instantaneous** acceleration is given by $x \to \frac{dx}{dt} \to v \to \frac{dv}{dt} \to a$

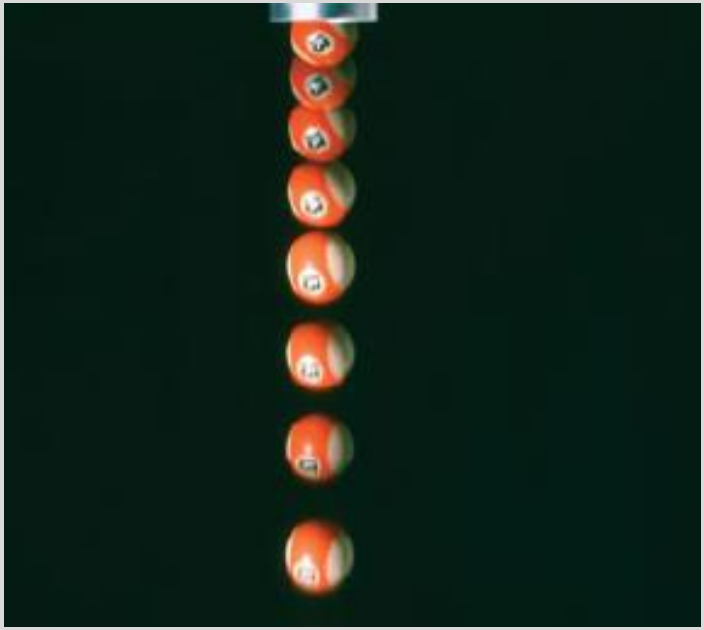
$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$



Motion with constant acceleration

Example

Many applications of mechanics involve objects moving with constant acceleration. For example: an object in free fall near Earth's surface.



Motion with constant acceleration

Velocity by algebra-based derivation

Goal: Given the acceleration is constant $a(t) = \text{const.}$

Derive expressions for the velocity $v(t) =?$ and the displacement $x(t) =?$

Derivation:

$$a = \frac{v_f - v_i}{t_f - t_i}$$

IF a is constant

= constant ⇒ $at = v - v_0$

$v = v_0 + at$

Choose the initial time $t_i = 0$ and t_f be any arbitrary time $t_f = t$.

Also, let $v_i = v_0$ (the initial velocity at $t = 0$) and $v_f = v$ (the initial velocity at any arbitrary time t?

Motion with constant acceleration

Velocity by algebra-based derivation

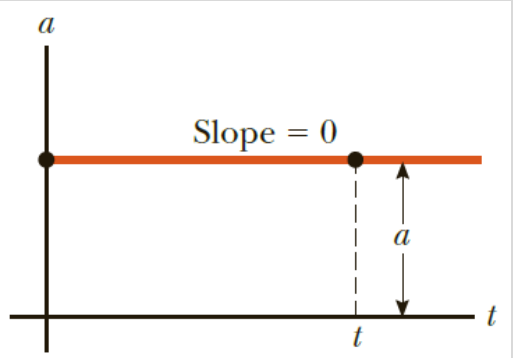
We have proved that

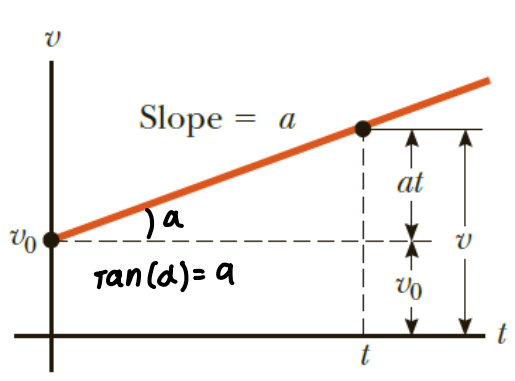
$$v = v_0 + at$$

Now, let's discuss the **graphical interpretation.**

Recall: the "slope" of velocity is the acceleration

$\tan(\alpha) = a = \frac{\text{oppo}}{\text{adja}} = \frac{at}{t}$





Motion with constant acceleration

Displacement by algebra-based derivation

Goal: Given the acceleration is constant $a(t) = \text{const}$.

Derive expressions for the velocity $v(t) =?$ and the displacement $x(t) =?$

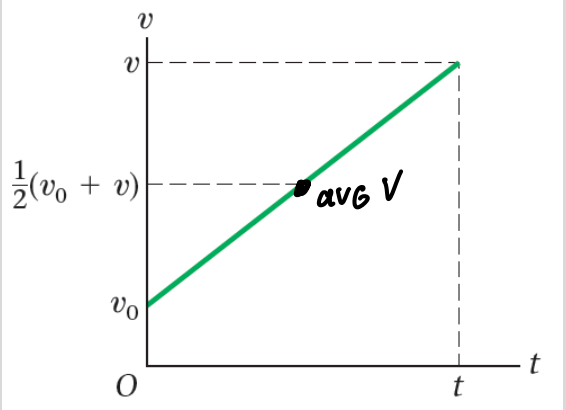
Derivation (cont): What about the dislacement $x(t)$?

The average velocity is $\bar{v} = (v_0 + v)/2$.

The displacement is given by

$$\Delta x = x_f - x_i = x - x_0 = \bar{v}\Delta t$$

$\bar{v} = \frac{\Delta x}{\Delta t}$ $\therefore$ $\Delta x = \bar{v}\Delta t$



$x = \bar{v}\Delta t + x_0$

$= \left(\frac{v_0 + v}{2}\right)\Delta t + x_0$

$= \left(\frac{v_0 + (v_0 + at)}{2}\right) t + x_0$

$= \frac{2v_0 + at}{2} t + x_0$

$= \frac{2v_0 t + at^2}{2} + x_0 = v_0 t + \frac{1}{2}at^2 + x_0$

Motion with constant acceleration

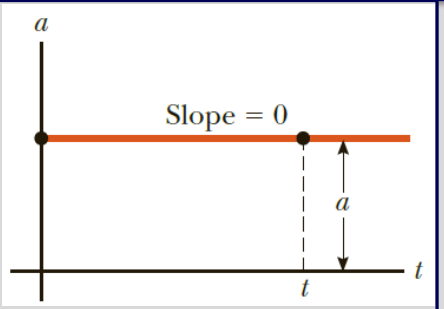
Equation	Information Given by Equation
$v = v_0 + at$ ✓	Velocity as a function of time
$\Delta x = v_0 t + \frac{1}{2}at^2$ ✓	Displacement as a function of time
$v^2 = v_0^2 + 2a\Delta x$	Velocity as a function of displacement

$\Delta x = v_0 t + \frac{1}{2}at^2 = v_0\left(\frac{v-v_0}{a}\right) + \frac{1}{2}a\left(\frac{v-v_0}{a}\right)^2$

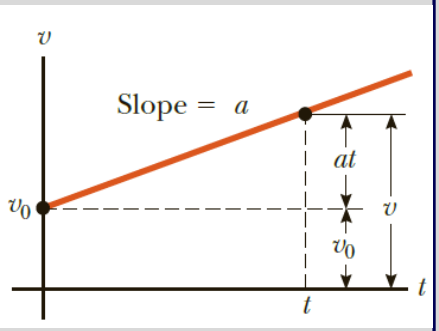
$\Rightarrow v^2 = v_0^2 + 2a\Delta x$

Linear: $y = ax + b$

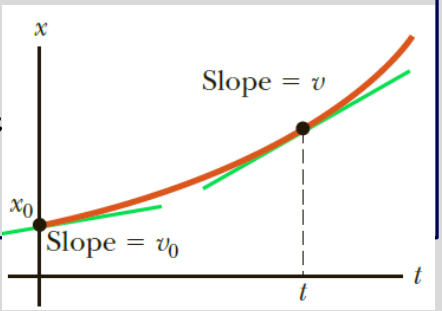
quadratic: $y = ax^2 + bx + c$



v is linear

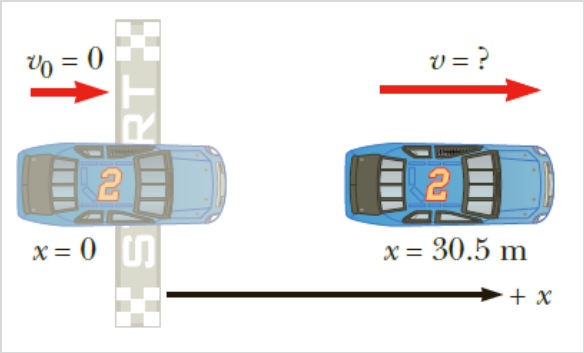


x is quadratic



Motion with constant acceleration

PROBLEM A race car starting from rest accelerates at a constant rate of 5.00 m/s^2. **(a)** What is the velocity of the car after it has traveled 1.00×10^2 ft? **(b)** How much time has elapsed? **(c)** Calculate the average velocity two different ways.



a) $v^2 = v_0^2 + 2a\Delta x$

$v^2 = 0 + 2(5)(30.5)$

$v^2 = 305$

$v = \sqrt{305} = 17.5$ m/s

b) $t = \frac{v - v_0}{a} = \frac{17.5}{5} = 3.5$ s

c) $\bar{v} = \frac{v + v_0}{2} = \frac{17.5}{2} = 8.75$ m/s

$\bar{v} = \frac{\Delta x}{t} = \frac{30.5}{3.5} = 8.71$ m/s

Motion with constant acceleration

Integration-based derivation

Equation	Information Given by Equation
$v = v_0 + at$	Velocity as a function of time
$\Delta x = v_0 t + \frac{1}{2}at^2$	Displacement as a function of time

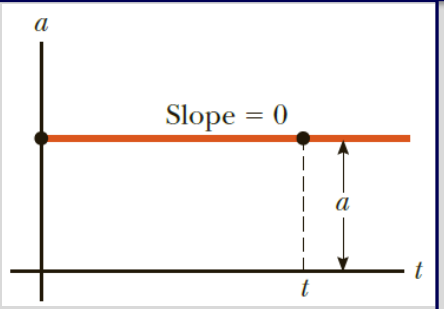
$a = \frac{dv}{dt} = \text{constant}$

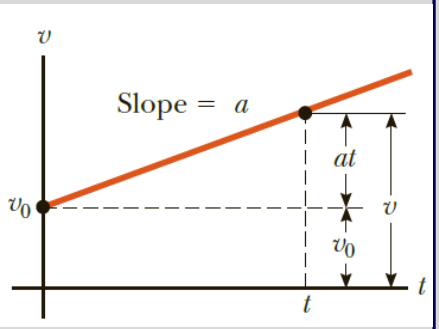
$\int_0^t dv = \int_0^t a\,dt \qquad \int dx = x \qquad (t=0)$

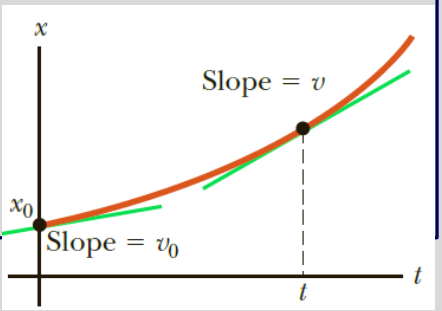
$\rightarrow v\int_0^t = a\int_0^t dt = at\Big|_0^t = at \rightarrow (t=0)$

$v - v(t=0) = at$

$v - v_0 = at \Rightarrow v = v_0 + at$







Free Falling Motion

Definition: Free falling moiton is the motion of an object moving under the influence of gravity only, and no other forces (**the air resistance is neglibible**).

Question: Does a heavier object fall faster than a light object?

NO Because gravity is constance

Free Falling Motion



Source Link.

Free Falling Motion

Question: Does a heavier object fall faster than a light object?

Answer:

True free-fall motion *does not* take object's **shape** or **mass** into account, i.e. a feather and a bowling ball will free-fall the same way.

Free Falling Motion

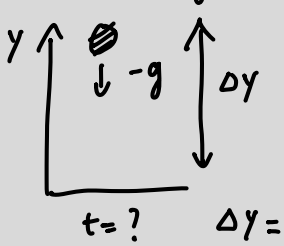
Kinematic equations

Equation	Information Given by Equation
$v = v_0 + at$	Velocity as a function of time
$\Delta x = v_0 t + \frac{1}{2}at^2$	Displacement as a function of time
$v^2 = v_0^2 + 2a\Delta x$	Velocity as a function of displacement

$a = -g \rightarrow v = v_0 - gt$

$\Delta y = v_0 t - \frac{1}{2} g t^2$

$v^2 = v_0^2 - 2g\Delta y$



$\Delta y = v_0 t - \frac{1}{2} g t^2 \rightarrow t = \sqrt{\frac{2\Delta y}{g}}$

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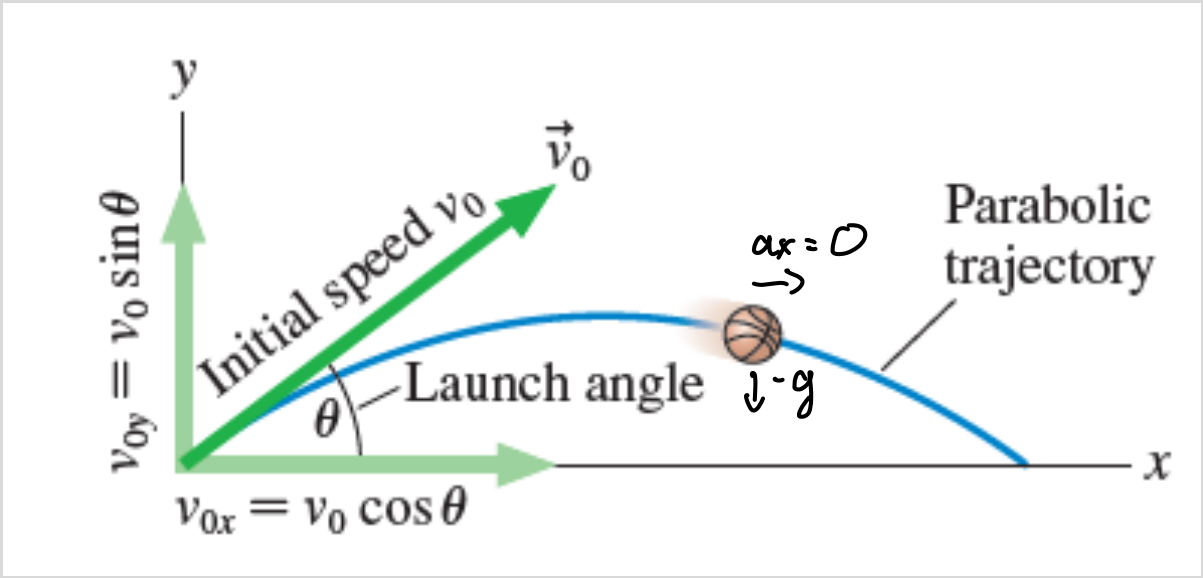
2. Motion in Two Dimensions

Motion on a plane.

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Projectile motion

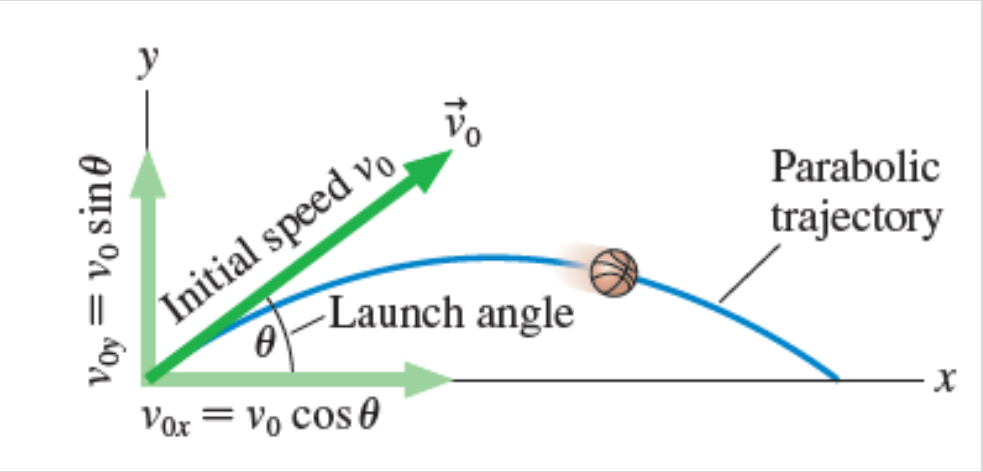
Introduction



Equation	Information Given by Equation
$v = v_0 + at$	Velocity as a function of time
$\Delta x = v_0 t + \frac{1}{2}at^2$	Displacement as a function of time
$v^2 = v_0^2 + 2a\Delta x$	Velocity as a function of displacement

Projectile motion

Kinematic equations



X direction

$v_x = v_{0x} + 0t = v_0 \cos\theta$

$\Delta x = (v_0 \cos\theta)t$

y direction

$v_y = v_{0y} + a_y t = v_0 \sin\theta - gt$

$\Delta y = (v_0 \sin\theta)t - \frac{1}{2} g t^2$

$v_y^2 = (v_0 \sin\theta)^2$

Projectile motion
Observations

In x-direction:

$$v_x = v_{0x} = v_0 \cos\theta_0 = \text{constant}$$

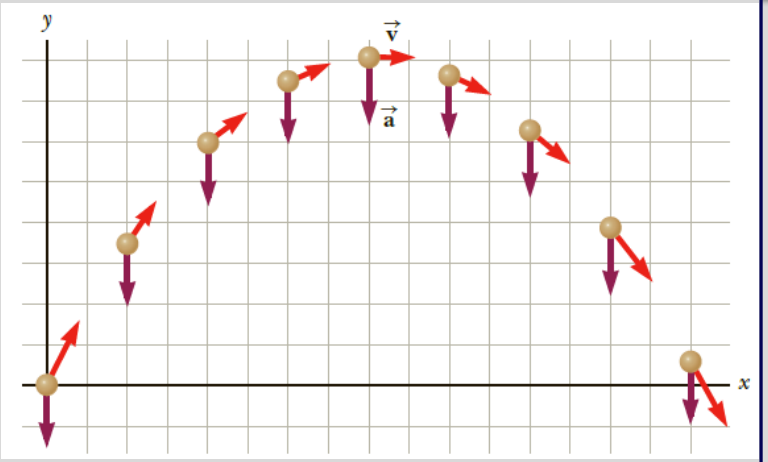
$$\Delta x = v_{0x} t = (v_0 \cos\theta_0)\, t$$

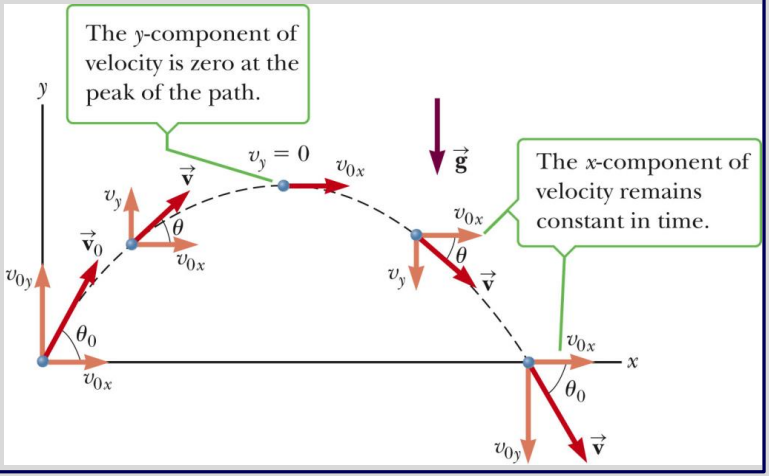
In y-direction:

$$v_y = v_0 \sin\theta_0 - gt$$

$$\Delta y = (v_0 \sin\theta_0)\, t - \tfrac{1}{2} g t^2$$

$$v_y^2 = (v_0 \sin\theta_0)^2 - 2g\,\Delta y$$

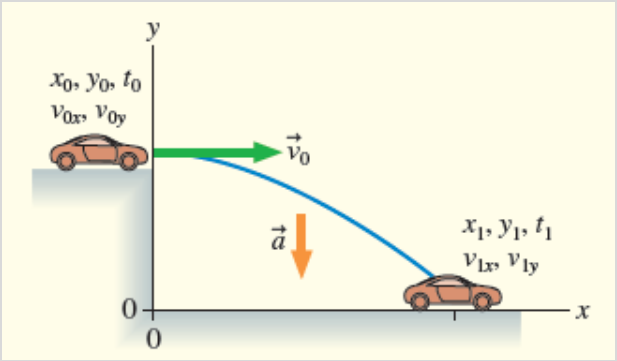




Projectile motion
Example

Example: A stunt man drives a car off a $10.0\ meters$ high cliff at a speed of $20.0\ m/s$ in the horizontal direction.

How far does the car land from the base of the cliff?



(1) Identify our initial velocity vector $\overrightarrow{v_0}$:

$$\overrightarrow{v_0} = [20, 0]$$

(2) Kinematic equations

$$\begin{cases} x(t) = x_0 + v_{0x}\Delta t = v_{0x}t \\ y(t) = y_0 + v_{0y}\Delta t - \frac{1}{2}g(\Delta t)^2 = y_0 - \frac{1}{2}gt^2 \end{cases}$$

(3) Let's identify what we know

$v_{0x} = 20\ m/s,$

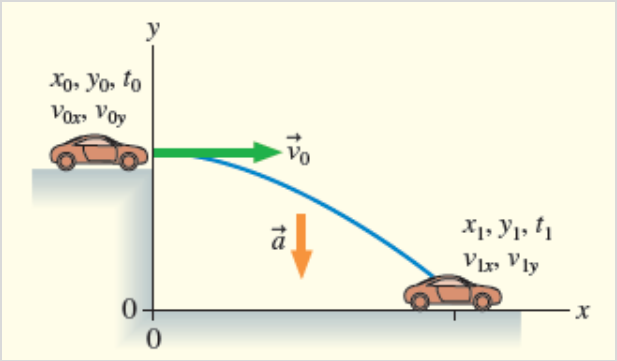
$y_0 = 10\ m,$

$g = 9.81\ m/s^2$

Projectile motion
Example

Example: A stunt man drives a car off a $10.0\ meters$ high cliff at a speed of $20.0\ m/s$ in the horizontal direction.

How far does the car land from the base of the cliff?



$$y(t) = 0 = 10 - \frac{1}{2}(9.81)t^2$$

$$\rightarrow t = \sqrt{10 \cdot \frac{2}{9.81}} = 1.43\ s$$

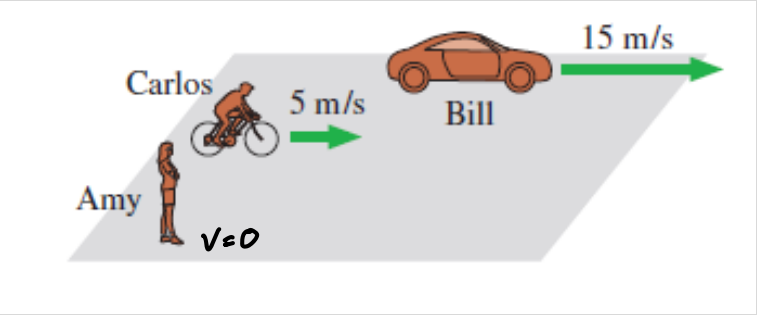
It takes the car 1.43 seconds to reaches the bottom of the cliff. In this same time interval, it would have traveled the following horizontal distance:

$$x(1.43) = 20\frac{m}{s} \cdot 1.43s = 28.6\ m.$$

Relative Motion
Introduction

- Amy was standing on the sidewalk when she saw her friends Carlos, who was riding a bicycle, and Bill who was driving his car.
  - According to Amy, Carlos was going at about $5\ m/s$. => $V_{Carlos\ Amy} = V_C - V_A = 5 - 0 = 5$
  - According to Bill, Carlos was moving backward at a velocity $-10\ m/s$.
- What is Bill's true velocity?

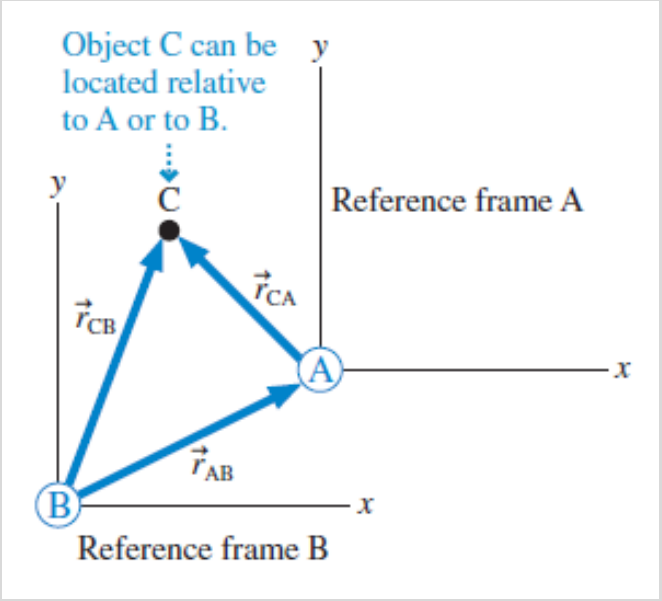
$V_{BC} = V_C - V_B = 5 - 15 = -10\ m/s$



Relative Motion

- The previous concept can be extended to **Two Dimensional problems** by establishing a **reference frame**.
- Consider 2 reference frames A and B and an object C. The position vector $\overrightarrow{r_{CB}}$ can be expressed as:

$$\overrightarrow{r_{CB}} = \overrightarrow{r_{CA}} + \overrightarrow{r_{AB}}$$



Relative Motion

- If we take the derivative of the previous equation with respective to time, we shall obtain:

$$\frac{d\overrightarrow{r_{CB}}}{dt} = \frac{d\overrightarrow{r_{CA}}}{dt} + \frac{d\overrightarrow{r_{AB}}}{dt}$$

$$\rightarrow \overrightarrow{v_{CB}} = \overrightarrow{v_{CA}} + \overrightarrow{v_{AB}}$$

- This equation was first discovered by Galileo so it is called **the Galilean transformation of velocity**.
- The equation indicates that if you know an object's velocity in one reference frame, you can transform it into the velocity that would be measured in a different reference frame.

Learning advising team (SAS)

Peter MOORE & JOSH REED WRITING

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2.4.40 engineering tutors

Learning Lab -> learn WRITING & math

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